Ensemble method	When to use it	Example use cases
Hard voting	Balanced classification dataset with multiple strong but diverse classifiers.	Spam detection, sentiment analysis, disease classification
Soft voting	Classification dataset with probabilistic models, where confidence scores matter.	Medical diagnosis, credit risk analysis, fake news detection
Bagging	Structured or semi-structured dataset with high variance and overfitting-prone models.	Financial risk modeling, ecommerce recommendation
Pasting	Structured or semi-structured dataset where more independent models are needed.	Customer segmentation, protein classification
Random forest	High-dimensional structured datasets with potentially noisy features.	Customer churn prediction, genetic data analysis, fraud detection
Extra-trees	Large structured datasets with many features, where speed is critical and reducing variance is important.	Real-time fraud detection, sensor data analysis
AdaBoost	Small to medium-sized, low-noise, structured datasets with weak learners (e.g., decision stumps), where interpretability is helpful.	Credit scoring, anomaly detection, predictive maintenance

Ensemble method	When to use it	Example use cases
Gradient boosting	Medium to large structured datasets where high predictive power is required, even at the cost of extra tuning.	Housing price prediction, risk assessment, demand forecasting